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Engineering Portfolio

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Two-Way Fluid-Structure Interaction Simulation

Overview

After the 2025-2026 SAE Aero Competition season completed, the team did a retrospective and determined that looking at planforms that are low-drag and high-velocity are the key to getting the most out of the ruleset, which was a stark departure from rulesets of years prior. To meet this demand, the team decided to look into potential thin-airfoil structures (<11% thickness) and alternative planforms, such as canards. After performing finite element analysis (FEA) on the derived structure of one of these airfoils, it was revealed that our standard manufacturing methods would cause a wing of 5ft half span to have a 4.4 inch deflection at the tip, significantly more than in prior years. To quantify further aerodynamic and structural impacts of this deflection, I developed a setup for FSI simulation between our team's validated tools of choice: Ansys Mechanical APDL and STAR-CCM+.

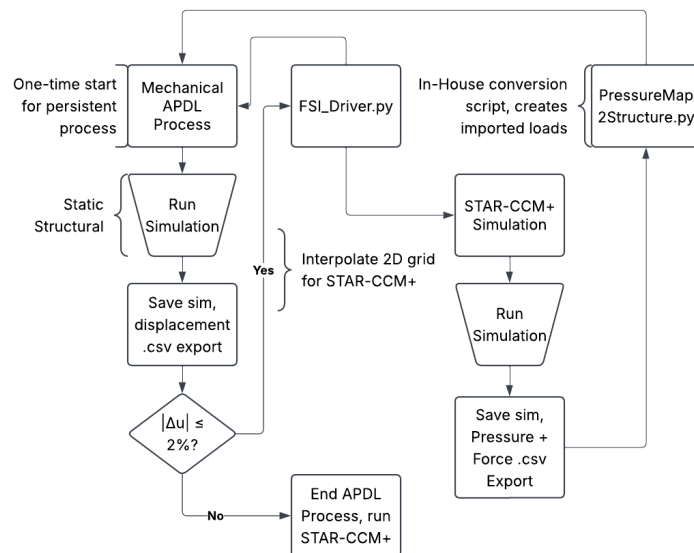
Since STAR-CCM+ doesn't have a direct API with Ansys, I created a Python script to take advantage of each program's automation features and have it run the process that would have been significantly more tedious by hand for more than one iteration of the loop.

Setup

A flowchart with the loop setup is seen to the right. FSI_Driver.py is the python script that the user runs, with displacement .csv exports being handled by APDL commands baked into the base simulation.

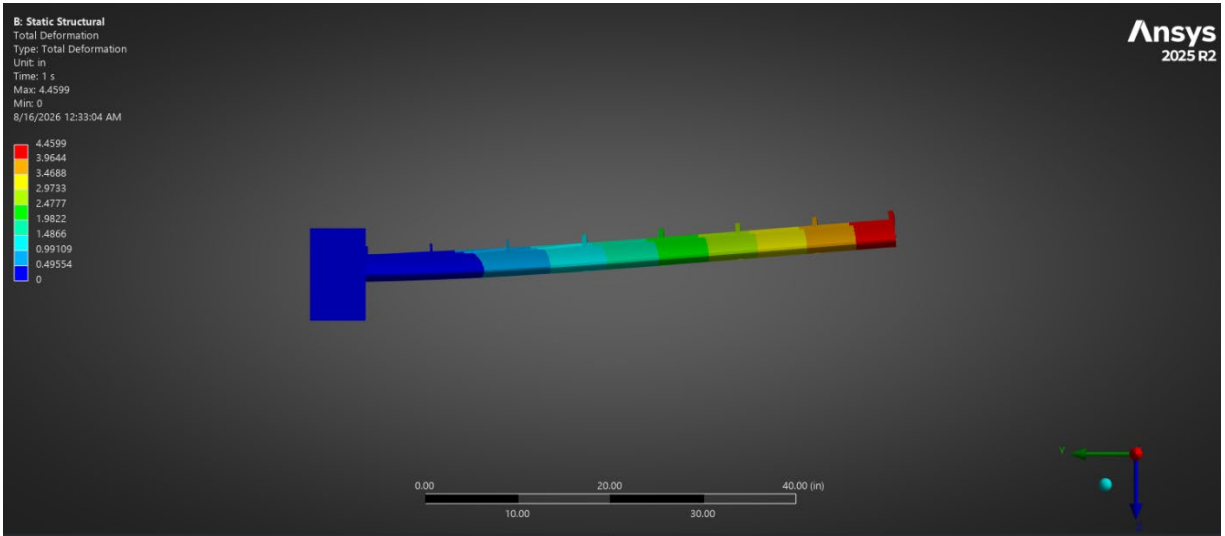
The STAR-CCM+ base sim was the same as the one the team traditionally uses but includes a simulation operation to deform the mesh according to the interpolated displacement table taken from Ansys.

The Ansys base sim is also identical to the one the team traditionally uses, outside of one extra APDL command snipped to export the displacements to a consistently-formatted .csv file.

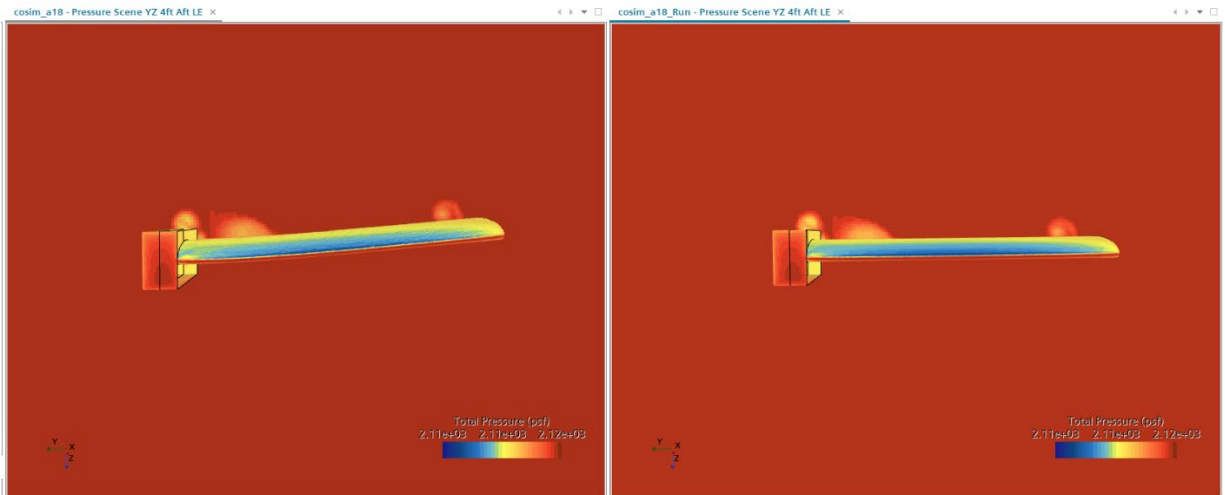


Results

An example load case of a 5ft half-span A18 airfoil with a 3ft chord was run, providing deflections and 2ft aft flow scenes as seen below.



Above: Total deflection scene of A18 wing structure as seen from the x-axis. Max deflection of 4.46 in encountered at wingtip trailing edge.



On Left: Pressure Scene of Deflected Structure. **On Right:** Pressure Scene of Static Structure
Both scenes taken using a YZ plane set 2ft aft of main wing trailing edge.

	Static Structure	Deflected Structure	% Difference
Lift (lbf)	47.387	46.830	-1.1%
Drag (lbf)	7.183	7.129	-0.75%
L/D	6.597	6.568	-0.44%

Challenges & Conclusion

One of the biggest challenges I ran into was learning Ansys's command syntax, as neither I nor anyone on the team have used APDL commands before (or had experience automating any part of Ansys's workflow), and Ansys's documentation was not the most user-friendly reference, so setting up the displacement export script was a challenge. Additionally, although STAR-CCM+ has a file-based cosimulation option using Ansys .cdb databases, that was for if you wanted to import that direct structure, and it would track the deformations of the Ansys structure. However, since the goal was to map the structural displacements directly to the outer mold line of the plane (the main difference between our Ansys and STAR-CCM+ simulations), the interpolation had to be done outside of the simulation. Therefore, the change in approach was made from directly using .cdb files to having a python script interpolate the exported displacements from Ansys across the entire outer mold line for STAR-CCM+ to deform using the simulation operations.

The resulting data shows that, with the designed structure, the impact of the structural deflection on aerodynamic numbers using the team's standard wing design is expected to be minimal, with the only area of concern being impacts on large horizontal stabilizers caused by the redirection of the vortex. On the plane last year, our horizontal stabilizer was significantly smaller span-wise than the main wing, so the impacts of this deflection on the horizontal stabilizer are minimal.

The next big step for the FSI loop is to determine whether it is possible and resource-friendly to simulate a structural deformation including the team's heat-shrink covering. This requires an initial thermal condition across a thin sheet of polyethylene, and requires more resources than initially expected. This portion of the investigation will continue once Ansys is functional on Texas A&M HPRC's systems, since the significance of the covering to the structural rigidity of the plane is still unquantified.

Propeller Modeling in STAR-CCM+

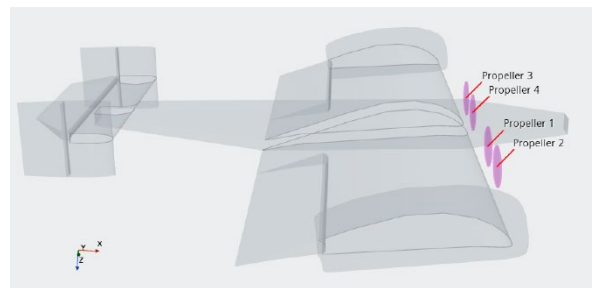
Overview

With a new ruleset arriving for Regular Class in the 2025-26 competition season, a new powertrain rule was brought about, meaning the team had a lot more freedom in determining what motors and propellers to use for the plane, along with motor placement. This was a big discussion in that competition year, as there was a debate between placing the motors right on the wing's leading edge and somewhere out in front, where they would be an independent structure (and easier for the structures team to design and build). Although this worked in practice and we were overall happy with the decision, questions remained on whether this had any aerodynamic impacts on the plane.

The team decided that determining how to simulate propeller wash onto the main wing would be something worth looking into, and we put it on the docket for something additional to validate, since we can test this in combination with developing an inertial measurement unit to validate our CFD simulations in real-world conditions, not just wind tunnels using a NACA 0012.

Setup

Using the team's standard STAR-CCM+ base simulation and base settings to maintain continuity between results, I set up the Virtual Disk model for all four propellers, using the Blade Element Method and standard propeller geometry taken from APC to set up the disks themselves. Rotation speed was determined by using ThrustLab's calculated motor RPM for the powertrain simulated. Blade Element Method was selected since it's a cheap way to get propeller impacts in a simulation, in comparison to modeling the full propeller in CAD and creating an overset mesh with an unsteady simulation, which is significantly more expensive resource-wise. An image of the setup can be seen to the right, with the four propeller disks of the simulated powertrain seen as the highlighted magenta. In a steady simulation, STAR-CCM+ permits a time-averaged calculation of propeller movement, which was chosen for resources. This model was designed with helicopter rotors in mind and was deemed suitable for our use case of propeller modeling.

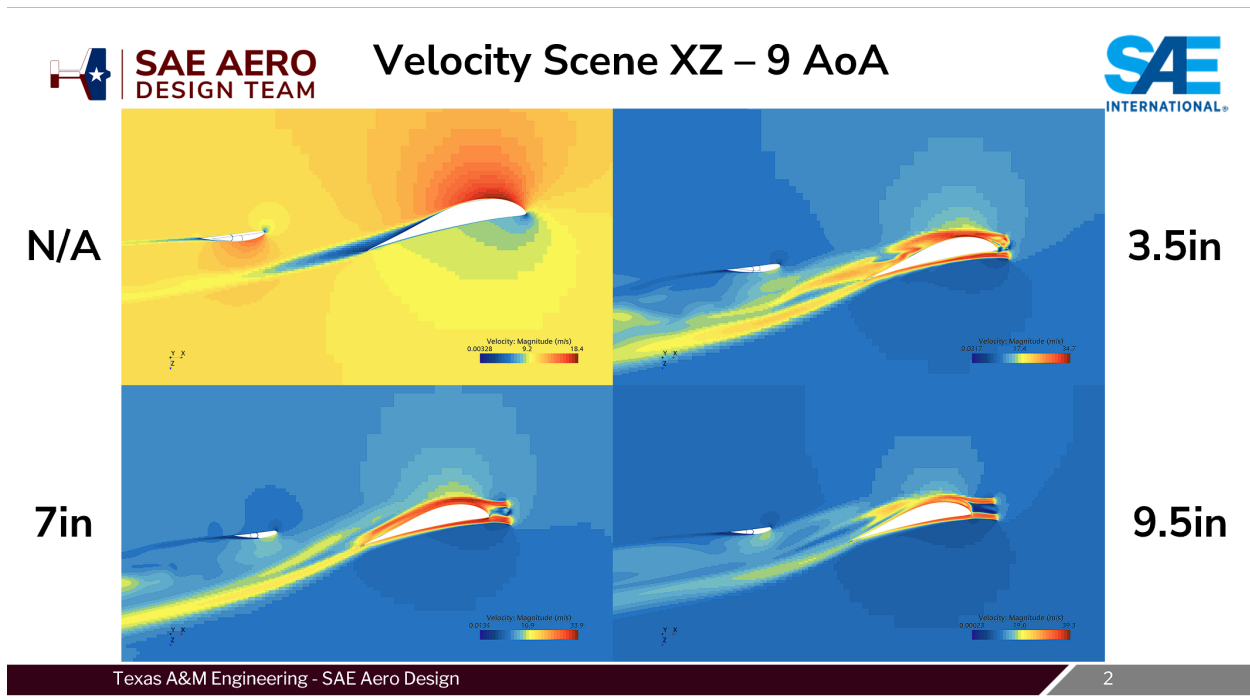
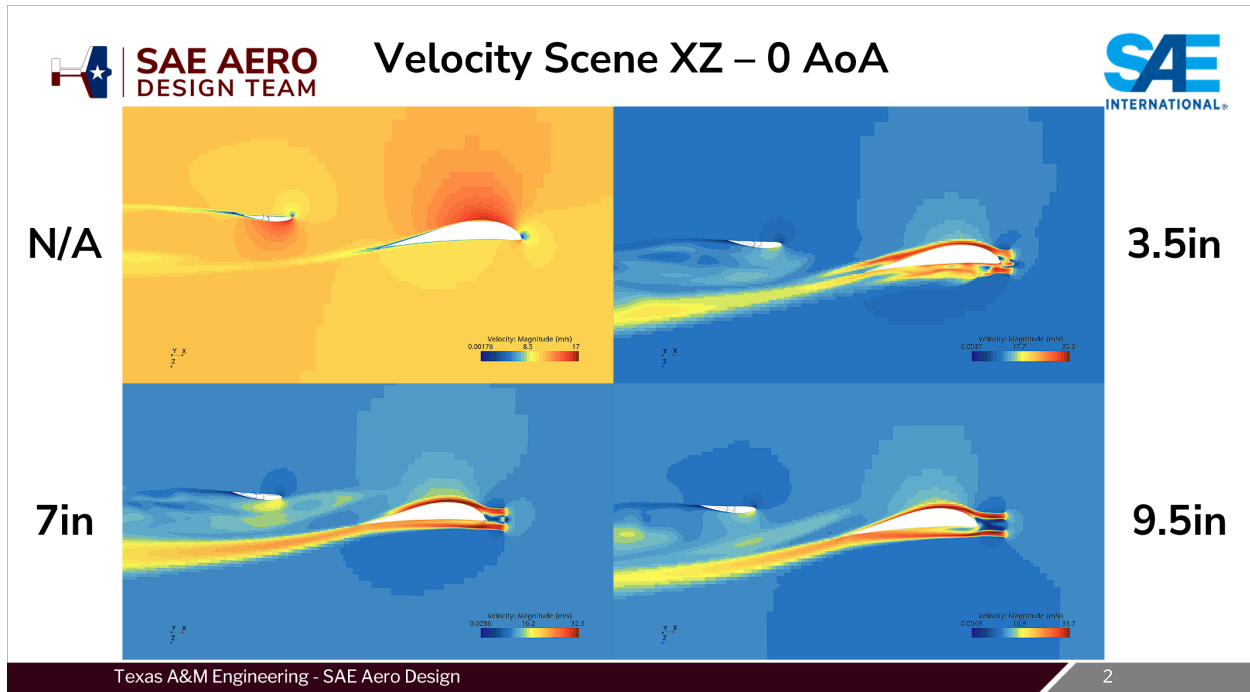


An identical setup was also run for an example biplane design, created by one of the two groups doing large-scale planform studies over the summer. Using various motor placements (2 ft gap on upper wing, 2 ft gap on lower wing, 1ft gap on upper wing, none) centered about the XZ midplane, impacts on lift, moment, and estimated trim AoA were analyzed.

Results

Model 1: 2025-2026 Competition Plane 1

The model was set up with our competition plane from the 2025-26 season, and had motors placed 3.5in, 7in, and 9.5in in front of the main wing leading edge, simulating wing LE motor placement, actual motor placement, and somewhere in between.



AoA	Main Wing Lift					H-Stab Downforce				
	0in	3.5in	7in	9.5in	None	0in	3.5in	7in	9.5in	None
0	43.30	42.94	43.79	43.57	35.44	6.23	7.02	6.69	6.70	5.34
3	53.48	54.35	55.90	55.44	46.03	5.90	6.44	6.69	6.75	4.88
6	64.55	66.48	67.50	66.91	56.59	5.38	6.73	6.83	6.49	4.37
9	75.23	78.66	79.67	78.16	66.90	5.25	6.08	6.51	7.05	3.85
12	87.80	90.46	91.19	89.65	70.31	5.22	6.09	6.62	6.86	1.70

Model 2: 2026-2027 Biplane Planform Study

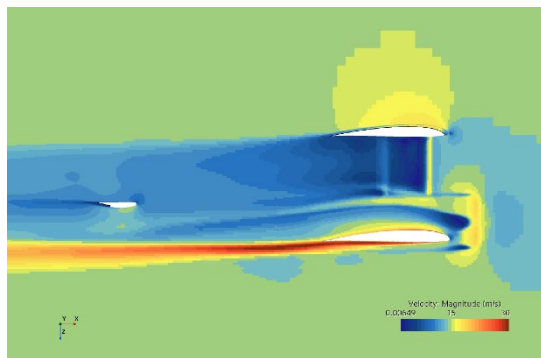


Figure 1.
Motors
placed on
lower
wing (left)

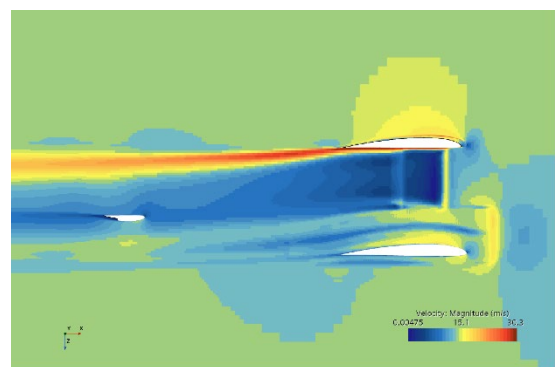


Figure 2.
Motors
placed on
upper
wing
(right)

AoA	L/D			
	2U	2L	1U	0
0	4.43	3.94	4.04	4.04
3	6.20	5.72	5.74	5.80
6	6.63	6.46	6.29	6.17
9	6.44	6.26	6.21	5.97
12	5.94	5.73	5.85	5.29

AoA	M Moment CG			
	2U	2L	1U	0
0	8.73	4.71	9.29	11.08
3	3.56	-0.94	4.11	5.87
6	-4.23	-8.29	-3.51	-1.08
9	-14.68	-18.63	-14.87	-8.76
12	-26.54	-27.91	-27.04	-17.61

Trim AoA @ 10% SM			
2U	2L	1U	0
5.066	2.500	5.386	6.384

Notes:

- U represents upper wing, L represents lower wing, numbers represent gap between motors in ft
- 0 represents a simulation with no motors
- L/D is presented for a more representative view of why motor placement matters more than expected in a biplane.
- Results of horizontal stabilizer downforce on the competition plane are much more significant than expected, it's believed to be a coincidence due to horizontal stabilizer z-position.
- Further validation is required to confirm the physical impact of these results.

GPU-Accelerated CFD Simulations

Overview

STAR-CCM+ has been slowly introducing GPU support for all its models from 2022 onwards, which is intended to significantly speed up simulation runtime and significantly reduce resource and power consumption per simulation. The team has run into significant issues in the past with running out of resources, since the version used was a few years old and before Siemens began adding GPU compatibility. To determine if it was worth swapping to a newer version and get the rest of the team up-to-speed with the new setup (as the UI had some significant differences to the incumbent version), I conducted a study on resource usage, simulation time, and resource availability using the current version and the new version of STAR-CCM+.

Setup

Using a base simulation of the 2024-2025 season's competition plane at 3 degrees AoA, meshes were generated at base sizes between 1m and 2m at equal increments, resulting in meshes ranging between 2 million and 40 million cells. Texas A&M HPRC's Grace supercomputer has nodes containing CPUs only, 2 A100s per node, 3 A40s per node, 2 RTX6000s per node, and 4 T4s per node, so simulations were set up utilizing all these nodes for study.

To confirm that no issues arise with setup across the new versions, one example simulation run was completed, ensuring all force, moment, and plane information reports were appropriately set up, and that variances of values between versions were under 2% across all relevant metrics.

Results

Configuration	Mesh Base Size (m)	Percent Savings vs. CPU v17			
		Elapsed Time	Total Time	SUs	SUs/Million Cells
A100 GPU - 1 Node - 2 GPUs Per Node	1	86.23%	86.27%	67.97%	68.91%
	1.125	80.37%	80.08%	54.32%	55.46%
	1.25	77.03%	76.92%	46.39%	47.38%
	1.375	83.37%	83.46%	61.20%	61.70%
	1.5	74.73%	74.50%	41.03%	41.55%
	1.625	65.82%	65.83%	20.24%	21.01%
	1.75	68.33%	68.00%	26.12%	27.31%
	1.875	59.89%	59.76%	6.39%	10.33%
	2	56.54%	57.47%	-1.43%	2.62%
Configuration	Mesh Base Size (m)	Percent Savings vs. CPU v17			
		Elapsed Time	Total Time	SUs	SUs/Million Cells
RTX 6000 GPU - 1 Nodes - 2 GPUs Per Node	1	72.69%	68.08%	54.64%	55.97%
	1.125	63.50%	53.02%	39.33%	40.84%
	1.25	62.26%	52.04%	35.88%	37.06%
	1.375	71.87%	71.72%	53.11%	53.72%
	1.5	57.82%	57.80%	29.71%	30.33%
	1.625	47.28%	47.69%	12.12%	12.97%
	1.75	48.60%	49.42%	14.35%	15.73%
	1.875	39.13%	40.24%	-1.48%	2.80%
	2	34.87%	34.51%	-8.53%	-4.22%
Configuration	Mesh Base Size (m)	Percent Savings vs. CPU v17			
		Elapsed Time	Total Time	SUs	SUs/Million Cells
T4 GPU - 1 Node - 4 GPUs Per Node	1	61.10%	60.93%	41.84%	43.55%
	1.125	59.80%	46.02%	39.87%	41.37%
	1.25	54.64%	12.04%	31.96%	33.21%
	1.375	72.32%	54.19%	58.47%	59.01%
	1.5	39.48%	-22.81%	9.23%	10.02%
	1.625	50.51%	8.84%	25.77%	26.49%
	1.75	47.76%	47.75%	21.63%	22.89%
	1.875	34.64%	34.55%	1.97%	6.09%
	2	35.12%	35.84%	2.68%	6.56%

Notes:

- Elapsed time represents true solver time; total time represents wall time according to the compute cluster.
- All GPU setups reduced true solver time, while most setups reduced SU (resource) usage.

Challenges & Conclusion

The big challenge that was encountered when ensuring the simulation was correctly set up in the new version is that documentation of changes between each version iteration is well documented, but

Based on the SU results, it was determined that as mesh cell size increases in the simulation, the more time and resources are saved by swapping over to GPU compute, assuming all other conditions are the same. Most of our simulations have similar mesh counts to the ones run at the 1m base size in the table above, meaning that guiding everyone to swap over to v19.06.008 of STAR-CCM+ from v17.02.008 was an easy decision to make. With later simulations, it was also determined that K-omega Shear Stress Transport simulations using the Implicit Unsteady solver in STAR-CCM+ would also run significantly faster and with much less resources than CPU compute, and opened up using that solver (along with other more-demanding solvers) to further analyze flow separation and stall regimes with our aircraft and figure out methods to either mitigate early stall or determine the stall wasn't severe enough at those angles of attack to worry about them.